

1. Simplex Method (Maximization Problem)

When to use: Objective is **Maximize Z** and Constraints are $\leq$ type

Steps:

1. Convert to standard form $\rightarrow$ add **slack variable** ($S \geq 0$)
2. Form initial simplex table
3. **Entering variable** $\rightarrow$ most negative in Z-row
4. **Leaving variable** $\rightarrow$ minimum ratio (RHS / positive entry)
5. Pivot $\rightarrow$ make column unit
6. Repeat until no negative in Z-row

No negative in $(Z_j - C_j)$ row $\rightarrow$ **Optimal solution reached**

2. Minimization Problem (Simplex)

When to use: Objective is **Minimize Z** and Constraints are $\geq$ or $=$ type

Case 1: Convert to Maximization

- Multiply objective by **-1** $\rightarrow$ it will convert $\geq$ to $\leq$
- Solve as maximization

Case 2: Dual Method (Shortcut)

- Convert primal $\rightarrow$ dual
- Solve as maximization

Shortcut: In exams, easiest $\rightarrow$ **convert to maximization**

3. Big-M Method

When to use: Constraints have $\geq$ or $=$

- Convert to standard form $\rightarrow$ subtract **slack variable** (S)
- Add **artificial variables** (A)
- Give them **very large penalty** (M) in objective function

4. Two-Phase Method

When to use: Same as Big-M ($\geq$ or $=$)

Phase 1 (Feasibility)

- Add artificial variables
- Minimize:

W = sum of artificial variables

Goal: make all artificial variables = 0

Phase 2 (Optimality)

- Remove artificial variables
- Use original objective function
- Continue simplex
- If $W=0 \rightarrow$ feasible
- If $W \neq 0 \rightarrow$ no solution

Simplex Method Chart

Method	constraints	Idea	Advantage
Maximization	$\leq$	Direct solve	Easy
Minimization	Min Z	Convert to Max	Simple trick
Big-M	$\geq, =$	Penalize artificial	Faster
Two-Phase	$\geq, =$	2-step solving	Safer

Transportation Problem

1. North-West Corner Rule (NWCR)

[Start from the **top-left corner (North-West)** and allocate.]

1. Start at cell (S1, D1)
2. Allocate = **min(Supply, Demand)**
3. Adjust supply & demand
4. Move:
 - Right $\rightarrow$ if demand satisfied
 - Down $\rightarrow$ if supply exhausted
5. Repeat until all filled

2. Least Cost Method (LCM)

[Always choose **minimum cost cell first**]

Steps:

1. Find smallest cost in table
2. Allocate max possible = min(supply, demand)
3. Cancel satisfied row/column
4. Repeat for next smallest cost

3. Vogel's Approximation Method (VAM)

[Use **penalty (difference of two smallest costs)**]

Find Penalty

- Row penalty = difference between 2 smallest costs in row
- Column penalty = same
- Choose **row/column with highest penalty**
- In that row/column → choose **minimum cost cell**
- Allocate max possible
- Adjust table and repeat

Transportation Problem Chart

Method	Idea
NWCR	Top-left start
LCM	Minimum cost
VAM	Penalty method